

Dilan Karaguler

Ph.D. Candidate in Mathematics, Michigan State University

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RESEARCH INTERESTS

Applied algebraic topology and topological data analysis; path homology and Mayer (N-nilpotent) homology of directed graphs and path complexes; persistent homology; topology-informed graph machine learning and geometric deep learning, with applications to molecular, dynamical and engineering data.

EDUCATION

Michigan State University

Ph.D. in Mathematics (Candidate)

– Advisors: Prof. Ekaterina Rapinchuk and Prof. Guowei Wei

East Lansing, MI, US

August 2022 – May 2027 (Expected)

Middle East Technical University (METU)

M.S. in Mathematics

– Thesis: *A Survey on Multidimensional Persistence Theory*

– Advisor: Assoc. Prof. Semra Pamuk

Ankara, Türkiye

September 2019 – September 2021

Middle East Technical University (METU)

B.S. in Mathematics, Honor Student

Ankara, Türkiye

June 2014 – June 2019

PUBLICATIONS

D. Karaguler and G.-W. Wei. *Mayer Path Homology*. Foundations of Data Science, accepted (2026).

doi:10.3934/fods.2026025, [arXiv:2605.16525](https://arxiv.org/abs/2605.16525).

D. Karaguler. *A Survey on Multidimensional Persistence Theory*. M.S. Thesis, Middle East Technical University (2021).

D. Karaguler and G.-W. Wei. *Persistent Mayer Path Laplacians*. In preparation (2026).

TALKS

Mayer Path Homology

TDA Seminar, Dept. of Computational Mathematics, Science and Engineering, MSU

April 2026

East Lansing, MI, US

Interaction Topology in TDA

Invited lecture, MTH 994: Mathematics-Assisted AI and AI-inspired Mathematics, MSU

October 2025

East Lansing, MI, US

Mayer Topology in TDA

Invited lecture, MTH 994: Mathematics-Assisted AI and AI-inspired Mathematics, MSU

October 2025

East Lansing, MI, US

From Data to Barcodes

Graduate Student Seminar, Middle East Technical University

April 2021

Ankara, Türkiye

POSTER PRESENTATIONS

SIAM Great Lakes Section Meeting

Illinois Institute of Technology

September 2025

Chicago, IL, US

Mathematics of Data, Dynamics, and Life Sciences Conference

University of California, Irvine

March 2025

Irvine, CA, US

RESEARCH EXPERIENCE

Graduate Researcher

Michigan State University

August 2022 – Present

East Lansing, MI, US

– Develop Mayer path homology, a homology theory for directed graphs built from N-nilpotent differentials on path complexes; published in *Foundations of Data Science* (with G.-W. Wei).

– Implement topological feature extraction (path complexes, Mayer homology and Laplacians, persistent homology) in Python with GUDHI, NumPy and PyTorch, and combine these features with graph neural networks.

Research Mentor

Topology RTG, Michigan State University

May 2023 – June 2023

East Lansing, MI, US

– Mentored an undergraduate researcher on a topological data analysis project: project design, methodology and research presentation.

INDUSTRY EXPERIENCE

Software Engineer Intern

May 2026 – August 2026

Intel Corporation

Oregon, US

- Built an end-to-end graph transformation pipeline in Python that extracts schematic topology directly from chip layout files, without pre-existing schematic metadata.
- Modeled low-level device instances as graph nodes and extracted a local subgraph around each instance to capture its connectivity.
- Developed graph aggregation and embedding routines producing a vector representation per device, used to classify components as analog, digital or fill devices.
- Worked with Design Yield and Manufacturability engineers to evaluate the prototype for automated structural recognition in layout verification.

Graduate Technical Intern

May 2025 – August 2025

Intel Corporation

Oregon, US

- Performed spatial pattern analysis on semiconductor chip layer images with KLayout and Python to identify and cluster recurring structures across layout datasets.
- Applied graph neural networks to search for and classify similar patterns across multi-layer designs.
- Partnered with cross-functional yield teams to bring ML-based pattern recognition into automated yield-analysis workflows.

PROJECTS

Persistent Mayer Homology with Dynamic Mode Decomposition

April 2026

Python, NumPy, SciPy, GUDHI

[GitHub](#)

- Built a reproducible pipeline combining Vietoris–Rips filtrations, Mayer homology and Dynamic Mode Decomposition (DMD) to track time-varying topological features.
- Computed Mayer Betti trajectories on synthetic 3D point dynamics; a rank-10 DMD model reconstructed a 500-frame dataset with about 3.6% relative error and supported multi-step forecasting.

Homology-Fed Neural Network Framework

November 2024

Python, PyTorch, GUDHI, NumPy

[GitHub](#)

- Wrote a Python library computing Mayer homology and path-complex Laplacians on graph-structured data.
- Used the resulting topological features for molecular property prediction on the QM9 dataset with graph neural networks.

Cellular Automata on Curved Surfaces

March 2024

Python, NumPy, Matplotlib

[GitHub](#)

- Simulated Conway's Game of Life on a square, torus and sphere with surface-specific neighbor rules, and analyzed how board size and initial density affect population persistence.

TEACHING EXPERIENCE

Lead Teaching Assistant

August 2023 – Present

Department of Mathematics, Michigan State University

East Lansing, MI, US

- Lead and mentor a team of graduate and undergraduate teaching assistants; observe in-person and online sections and give constructive feedback.
- Currently lead TA for first-year graduate students; design and run workshops that prepare them for their first teaching assignments.
- Work with faculty to set best practices for teaching assistants and prepare teaching materials for multi-section courses.

Graduate Lead, Center for Instructional Mentoring (CIM)

Summer 2024

Department of Mathematics, Michigan State University

East Lansing, MI, US

- Mentored first-time instructors through non-evaluative teaching feedback, one-on-one conversations and ongoing guidance during their first teaching experience.
- Created a teaching handbook built on interviews with senior TAs about the courses they taught, describing each course's expectations so junior graduate students can choose assignments that fit their needs.

Mentor, Academic Culture and English Skills (ACES) Program

Summer 2024

Michigan State University

East Lansing, MI, US

- Mentored incoming international graduate students preparing to become MSU teaching assistants, introducing U.S. classroom culture, university structure and cross-cultural communication, and organized social events.

Supervisor, Math Learning Center

August 2022 – May 2023

Michigan State University (Teaching Exemption Fellowship)

East Lansing, MI, US

Graduate Teaching Assistant	January 2020 – June 2022
Middle East Technical University	Ankara, Türkiye
– Led recitations and office hours, graded, and proctored online and in-person exams for Calculus I–II and Differential Equations.	
Undergraduate Teaching Assistant	March 2018 – June 2019
Middle East Technical University	Ankara, Türkiye
– Graded homework and held review hours for Calculus for Mathematics Students.	

COURSES TAUGHT

MSU MTH 299 — Transitions	Fall 2025
MSU MTH 314 — Matrix Algebra with Computational Applications	Spring 2025
MSU MTH 235 — Differential Equations	Spring 2024
MSU MTH 133 — Calculus II	Summer 2023, Fall 2023, Fall 2024
METU MTH 154 — Calculus II for Mathematics Students	Spring 2022
METU MTH 153 — Calculus I for Mathematics Students	Fall 2021
METU MTH 120 — Calculus II for Engineering Students	Spring 2021
METU MTH 119 — Calculus I for Engineering Students	Fall 2020

AWARDS, FELLOWSHIPS & SCHOLARSHIPS

Doctoral Student Summer Internship Fellowship, The Graduate School, Michigan State University	Summer 2026
Excellence in Teaching Award , Department of Mathematics, Michigan State University	May 2025
Colleges Online Learning Academy (COLA) Fellowship, Michigan State University	Summer 2024
Teaching Exemption Fellowship, Michigan State University	2022 – 2023
RONESANS Scholarship, RONESANS	2018 – 2019
UNIGENC Scholarship, UNIGENC Association	2015 – 2019
Merit Scholarship, Turkish Ministry of National Education	2010 – 2014

SERVICE & LEADERSHIP

Graduate Student Representative, Graduate Studies Committee	2025 – 2026
Department of Mathematics, Michigan State University	East Lansing, MI, US
– Represented all mathematics graduate students at committee meetings with faculty.	
– Planned and prepared Responsible Conduct of Research (RCR) workshops with the Graduate Chair.	
– Met with prospective graduate students and hosted graduate student town hall meetings on the state of the department.	
– Served as a point of contact for graduate students with grievances and directed them to department and university resources.	
President / Board Member	August 2022 – August 2025
Association for Women in Mathematics (AWM), MSU Chapter	East Lansing, MI, US
– Organized seminars, career workshops, informational sessions on academic life and social events for women in mathematics.	
Organizing Committee Member	April 2024
Graduate Student Topology and Geometry Conference, Michigan State University	East Lansing, MI, US
– Handled communication with speakers and attendees, room assignments and talk scheduling, and moderated sessions.	
President	February 2015 – December 2018
METU Math Club	Ankara, Türkiye
– Organized seminars, workshops and regular reading groups in topology, algebra and other areas.	

TECHNICAL SKILLS

Programming: Python, R, SQL

Methods: Graph Neural Networks, Graph Transformers, Topological Data Analysis, Persistent Homology, Vietoris–Rips Filtrations, Dynamic Mode Decomposition, Supervised/Unsupervised Learning, Model Evaluation

Libraries: PyTorch, PyTorch Geometric, NetworkX, GUDHI, NumPy, SciPy, Pandas, Scikit-Learn, TensorFlow, Keras, Matplotlib, XGBoost, LightGBM

Tools: Git, Linux/Unix, DuckDB, SQLite, Jupyter, KLayout, L^AT_EX

Domain Knowledge: Design for Manufacturability (DFM), Semiconductor Yield Analysis, Schematic Extraction, Chip Layout Analysis

REFERENCES

Available upon request.